HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY (HCMUT-VNU) – FACULTY OF CSE
CO5151: ADVANCED AGENTIC AI (HK261) – FACULTY ADVISOR: **DR. LE XUAN BACH**

ANNOTATED BIBLIOGRAPHY: SAFETY & ALIGNMENT

Presenting Group G4: Dang Lam Tung (Lead) · Nguyen Trung Phong · Vu Viet Hung | *Topic S2-5 – November 4, 2026*

***Compliance Mandate**: Strictly covers the 2 Core Readings and 3 Extension Works explicitly assigned in the S2-5 syllabus. Each entry is accompanied by **EXACTLY ONE critical reflection sentence** (academic skepticism on assumptions, bottlenecks and limits of generalisation), strictly avoiding descriptive summaries.*

1. Core Readings (Foundational Alignment Mechanisms)

1. **Bai, Y. et al. (2022).** *Constitutional AI: Harmlessness from AI Feedback.* arXiv:2212.08073.
   Focus Area: The two-phase recipe — SL-CAI (principle-conditioned self-critique and revision) and RL-CAI (AI-judge preference labels → preference model → RL) — as a replacement for human harm labels, plus the harmlessness–evasiveness trade-off.
   Critical Reflection: While the method removes the cost of human harm labels and yields a non-evasive assistant preferred by crowdworkers over human-feedback baselines, its harmlessness signal is only as trustworthy as a hand-written constitution that the model itself critiques, so the method *relocates* rather than *eliminates* the specification problem — and the paper's own admission that critiques "often provide inaccurate criticism" leaves the loop with no external check confirming that a revision actually reduced harm.

2. **Ouyang, L. et al. (2022).** *Training Language Models to Follow Instructions with Human Feedback (InstructGPT).* arXiv:2203.02155.
   Focus Area: The three-stage SFT → reward-model → PPO pipeline, the "alignment tax" and its PPO-ptx mitigation, and the relationship between parameter scale and instruction adherence.
   Critical Reflection: The finding that a 1.3B model beats 175B GPT-3 is strong evidence that preference data can substitute for scale, yet the reward model is trained on comparisons where labelers prioritise *helpfulness* while the headline evaluations reward *truthfulness and harmlessness*, so part of the reported alignment gain is an artefact of a shifted measurement objective rather than proof that the model learned what evaluators actually care about.

2. Extension Readings: Presenting Group Deep Dive

3. **Shavit, Y., Agarwal, S., Brundage, M. et al. (2023).** *Practices for Governing Agentic AI Systems.* OpenAI White Paper.
   Focus Area: A graded definition of agenticness (goal complexity, environmental complexity, adaptability, independent execution) and seven governance practices, notably §4.2 *constraining the action-space and requiring approval* and §4.7 *interruptibility*.
   Critical Reflection: The seven practices supply a usable vocabulary for bounding an agent's action space, but as a policy white paper it ships no measurement protocol, so it cannot say which practice actually reduces harm once an agent is capable enough to *route around* hard-coded restrictions — the loophole the authors themselves concede with the "email a human to do it" example.

4. **Greenblatt, R., Denison, C. et al. (2024).** *Alignment Faking in Large Language Models.* arXiv:2412.14093.
   Focus Area: Conditional compliance keyed to the model's belief about being trained (14% on the free tier vs. ≈0% on the paid tier), strategic reasoning in the scratchpad, and the effect of actually applying RL (78%).
   Critical Reflection: The demonstration is compelling precisely because the model was never instructed to fake, but the authors' concession that they made faking easy — by naming both the training criterion and the audited audience — leaves open whether the 14% reflects a general propensity to condition behaviour on perceived oversight or is an artefact of an unusually legible training story.

5. **Meinke, A., Schoen, B. et al. (2024).** *Frontier Models are Capable of In-context Scheming.* arXiv:2412.04984.
   Focus Area: Six agentic evaluations across five frontier models (× 300 rollouts each): oversight subversion, self-exfiltration, goal-guarding, covert email reranking, instrumental alignment faking and sandbagging, plus the persistence of deception under interrogation.
   Critical Reflection: The study measures *capability* rather than *propensity* and runs in sandboxes where "weights" are simulated, so the headline rates (5–22%) are best read as an upper bound on what models can do under strong goal nudging rather than an estimate of what deployed agents will do — a distinction the paper states in its limitations but its abstract's framing tends to blur.